The class of unconfined systems has the bounded amalgamation property with an explicit bound

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Editorial note. The counted statement is the author's environment at source0.tex lines 740-741 and its proof environment opens at line 742; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial note. r002 note: r001 failed because the retained author proof cites two bibliography keys of the article; r002 bundles the article's own bibliography file and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: the bounded amalgamation property with an explicit bound for unconfined systems (author label Feb_23_2024_2).

Allowed standard prerequisites (editorial).

- Amalgamation in a bound. A$\leq$n B means that the extension absorbs any set of at most n new elements.. It is the bound that the proposition makes explicit.

- Two HF-orderings combined. An HF-ordering of C can be concatenated with the restriction of an HF-ordering of B* to the complement of C to give an HF-ordering of the union.. It is the construction that produces the amalgam.
- Isomorphic copy over A. B* is a copy of B fixing A, arranged so that the union with C is unconfined.. It is the choice made at the start of the proof.

Lemma 0.1 1. Assume $A \leq B$, $b \in B \setminus A$ and there is a block $\{a_1, a_2, b\}$ with $a_1, a_2 \in A$. Then $Ab \leq B$.

2. Assume $A \leq^n B$, $b \in B \setminus A$ and there is a block $\{a_1, a_2, b\}$ with $a_1, a_2 \in A$. Then $Ab \leq^{n-1} B$.

Proposition 0.2 $(\mathcal{C}_{\rm un}, \leq)$ has the n-amalgamation property for every n. In fact, if $A \leq B$ and $n < \omega$ are given, we can always take $m = n + |B \setminus A|$.

Proof. Assume $A \leq B$ and $A \leq^m C$, where $m = n + |B \setminus A|$. We want to find $B^* \cong_A B$ and $D \in \mathcal{C}_{\rm un}$ such that $C \leq D$ and $B^* \leq^n D$. We may assume $C \cap B = A$. Note that the blocks containing only elements of A are the same in B and in C. Assume there are $b \in B \setminus A$, $c \in C \setminus A$ and $a_1, a_2 \in A$ such that $\{a_1, a_2, b\}$ is a block of B and $\{a_1, a_2, c\}$ is a block of C. Redefine $A' = A \cup \{c\}$, $B' = (B \setminus \{b\}) \cup \{c\}$ and modify the blocks of B by exchanging b with c to obtain the blocks of B'. Then $B' \cong_A B$ and $B' \cap C = A'$. By Lemma 0.1, we have $A' \leq B'$ and $A' \leq^{m-1} C$. Iterating this process gives $B^* \cong_A B$ such that if $A^* = B^* \cap C$, then $A^* \leq B^*$, $A^* \leq^{n+i} C$ for some $i \geq 0$ and there are no $b \in B^* \setminus A^*$, $c \in C \setminus A^*$ and $a_1, a_2 \in A^*$ such that $\{a_1, a_2, b\}$ is a block of B^* and $\{a_1, a_2, c\}$ is a block of C. Now consider $D = B^* \cup C$ as a free amalgam of B^* and C over A^*, that is, with the blocks in B^* and in C and no others. Then D is a partial STS. We claim that $D \in \mathcal{C}_{\rm un}$, $C \leq D$ and $B^* \leq^n D$.

First we show that D has an HF-ordering where C is an initial segment. This proves that $D \in \mathcal{C}_{\rm un}$ and $C \leq D$. Take an HF-ordering $<_C$ of C and an HF-ordering $<_{B^*}$ of $B^* = A^* \cup (B^* \setminus C)$ where A^* is an initial segment. Then the sum

$$(C, <_C) + (B^* \setminus C, <_{B^*} \restriction (B^* \setminus C))$$

is an HF-ordering of D. If not, there is an element of $B^* \setminus C$ that belongs to different blocks containing smaller elements, but these elements then are in $A^* \cup (B^* \setminus C)$, contradicting the choice of the ordering $<_{B^*}$ of B^* as an HF-ordering over A^*.

Finally, we check that $B^* \leq^n D$. Let $X \subseteq D \setminus B^* = C \setminus A^*$ with $|X| \leq n$; we show that $B^* \leq B^* \cup X$. Since $A^* \leq B^*$, we have an HF-ordering $<_{B^*}$ of B^* with initial segment A^*. Since $A^* \leq^n C$, we know that $A^* \leq A^* \cup X$, and hence there is an HF-ordering $<_{A^* \cup X}$ of $A^* \cup X$ with initial segment A^*. We combine these orderings by taking the sum $(B^*, <_{B^*}) + (X, <_{A^* \cup X} \restriction X)$. Since an element of X cannot have a block containing elements of B^* unless the elements belong to A^*, we obtain an HF-ordering of $B^* \cup X$ with initial segment B^*. □

References

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